

Algebra

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Equalities

Mastery of [Elementary Algebra](#) (a.k.a. “College Algebra”) is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

Theorem 0.1 (Equalities are transitive). *If $a = b$ and $b = c$, then $a = c$*

Theorem 0.2 (Substituting equivalent expressions). *If $a = b$, then for any function $f(x)$, $f(a) = f(b)$*

Inequalities

Theorem 0.3 (Adding to both sides of an inequality). *If $a < b$, then $a + c < b + c$*

Theorem 0.4 (negating both sides of an inequality). *If $a < b$, then: $-a > -b$*

Theorem 0.5 (Multiplying both sides of an inequality by a positive number). *If $a < b$ and $c > 0$, then $ca < cb$.*

Theorem 0.6 (Negation is multiplication by -1).

$$-a = (-1) * a$$

Minimum, maximum, argmin and argmax

Exercise 0.1 (Smallest value, and where it occurs).

1. Let $f(x) = (x - 2)^2 + 1$ for every $x \in \mathbb{R}$. What is the smallest value f takes, and at which x does it take that value?
2. Now let $g(x) = (x - 2)^2 + 1$ only for $x \in \{0, 1, 3, 4\}$. What is the smallest value g takes, and at which x does it take that value?

Solution

Solution 0.1.

1. The square of a real number is never negative, so $(x - 2)^2 \geq 0$ for every x , and therefore $f(x) \geq 0 + 1 = 1$. The square is 0 only when $x - 2 = 0$, that is, when $x = 2$, and there $f(2) = 1$. So the smallest value of f is 1, and $x = 2$ is the only input where f takes it.
2. Evaluate g at each of its four inputs:

x	0	1	3	4
$g(x)$	5	2	2	5

The smallest value of g is 2, and g takes it at two inputs, $x = 1$ and $x = 3$. The answer to “where?” is the set $\{1, 3\}$, not a single number.

Definition 0.1 (Minimum). Let $A \subseteq \mathbb{R}$. A number m is the **minimum** of A , written $\min A$, if $m \in A$ and $m \leq a$ for all $a \in A$.

Not every set has a minimum. The interval $(0, 1]$ has none: every element a of it has a smaller element, $a/2$, also in it. In Exercise 0.1, the smallest value of g is $\min\{5, 2, 2, 5\} = 2$.

Definition 0.2 (Maximum). Let $A \subseteq \mathbb{R}$. A number M is the **maximum** of A , written $\max A$, if $M \in A$ and $a \leq M$ for all $a \in A$.

Definition 0.3 (Argmin). Let $f : A \rightarrow \mathbb{R}$ be a **function**. The **argmin** of f over A is the set of inputs where f takes its smallest value:

$$\arg \min_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A, f(x) \leq f(a)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg \min_{x \in A} f(x)$.

The smallest value itself is $\min f(A)$, the minimum (Definition 0.1) of the **image** of f , and the argmin is where that value is attained. In Exercise 0.1, $\arg \min_{x \in \mathbb{R}} f(x) = 2$ and $\arg \min_{x \in \{0,1,3,4\}} g(x) = \{1, 3\}$. The argmin is empty when f has no smallest value, for example $f(x) = x$ on $(0, 1]$.

Definition 0.4 (Argmax). Let $f : A \rightarrow \mathbb{R}$ be a **function**. The **argmax** of f over A is the set of inputs where f takes its largest value:

$$\arg \max_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A, f(a) \leq f(x)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg \max_{x \in A} f(x)$.

Global and local minimizers

Exercise 0.2 (A dip that is not the bottom). Let $f(x) = x^3 - 3x$ for $x \in \mathbb{R}$.

1. Show that $f(x) - f(1) = (x - 1)^2(x + 2)$.
2. Use part 1 to show that $f(x) \geq f(1)$ for every x in the interval $(0, 2)$.
3. Compute $f(-3)$. Is $f(1)$ the smallest value f takes on all of $\mathbb{R}$?
4. Does f take a smallest value anywhere on $\mathbb{R}$?

Solution

Solution 0.2.

1. Since $f(1) = 1 - 3 = -2$,

$$\begin{aligned} f(x) - f(1) &= x^3 - 3x + 2 \\ &= (x - 1)(x^2 + x - 2) \\ &= (x - 1)(x - 1)(x + 2) \\ &= (x - 1)^2(x + 2). \end{aligned}$$

2. For $x \in (0, 2)$, $(x - 1)^2 \geq 0$ and $x + 2 > 0$, so their product is at least 0. By part 1, $f(x) - f(1) \geq 0$, that is, $f(x) \geq f(1)$.
3. $f(-3) = -27 + 9 = -18$. Since $-18 < -2 = f(1)$, $f(1)$ is not the smallest value of f on $\mathbb{R}$.

4. No. For each $n \geq 2$, $f(-n) = -n^3 + 3n = -n(n^2 - 3) \leq -n$, so no value of f is smaller than all the others: for any x , picking $n \geq 2$ with $-n < f(x)$ gives $f(-n) < f(x)$.

Definition 0.5 (Global minimizer). Let $A \subseteq \mathbb{R}^p$ and let $f : A \rightarrow \mathbb{R}$ be a [function](#). A point $x^* \in A$ is a **global minimizer** of f over A if $f(x^*) \leq f(x)$ for all $x \in A$.

The global minimizers of f are exactly the elements of $\arg \min_{x \in A} f(x)$ (Definition 0.3). In Exercise 0.2, f has no global minimizer over $\mathbb{R}$.

Definition 0.6 (Local minimizer). Let $A \subseteq \mathbb{R}^p$ and let $f : A \rightarrow \mathbb{R}$ be a [function](#). A point $x^* \in A$ is a **local minimizer** of f if there is a number $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in A$ with $\|x - x^*\| < \delta$, where $\|\cdot\|$ is the [Euclidean norm](#).

For $p = 1$, $\|x - x^*\| = |x - x^*|$. Every global minimizer (Definition 0.5) is a local minimizer: take any $\delta > 0$. The converse fails. In Exercise 0.2, $x^* = 1$ is a local minimizer of f (take $\delta = 1$), but not a global one.

Convex functions

Exercise 0.3 (A chord above a parabola). Let $f(x) = (x - 2)^2$, and take the points $x = 0$ and $y = 4$ with $t = \frac{1}{2}$.

1. Compute $f(tx + (1 - t)y)$.
2. Compute $tf(x) + (1 - t)f(y)$.
3. Is $f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$?

Solution

Solution 0.3.

1. With $t = \frac{1}{2}$, $tx + (1 - t)y = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 4 = 2$, so $f(2) = (2 - 2)^2 = 0$.
2. $f(0) = (0 - 2)^2 = 4$ and $f(4) = (4 - 2)^2 = 4$, so $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 4 = 4$.
3. Yes: $0 \leq 4$. The value of f at the midpoint of 0 and 4 is below the average of its values at the two endpoints.

Definition 0.7 (Convex function). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be a **function**. f is **convex** if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all $x, y \in \mathbb{R}^p$ and all $t \in [0, 1]$.

The point $tx + (1-t)y$ lies on the line segment from x to y , and the right-hand side is the height of the chord joining $(x, f(x))$ and $(y, f(y))$ above that point. So f is convex when every chord lies on or above the graph. In Exercise 0.3, $f(x) = (x-2)^2$ with $x = 0$, $y = 4$, $t = \frac{1}{2}$ gives $f(2) = 0 \leq 4$, which is one instance of the inequality. The function in Exercise 0.2, $f(x) = x^3 - 3x$, is not convex. Take $x = -2$, $y = 0$, $t = \frac{1}{2}$: $f(-1) = 2$, but $\frac{1}{2}f(-2) + \frac{1}{2}f(0) = \frac{1}{2} \cdot (-2) + 0 = -1$, and $2 \leq -1$ is false.

Theorem 0.7 (Local minimizers of convex functions are global). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be a **convex function**. Every **local minimizer** of f is a **global minimizer** of f over $\mathbb{R}^p$.

For example, $f(x) = (x-2)^2$ is convex (by expanding the square, $f(tx + (1-t)y) - tf(x) - (1-t)f(y) = -t(1-t)(x-y)^2 \leq 0$), and $x^* = 2$ is a local minimizer of f . The theorem says $x^* = 2$ is a global minimizer, which is also visible directly: $f(x) = (x-2)^2 \geq 0 = f(2)$ for every x . The theorem fails without convexity: $f(x) = x^3 - 3x$ has a local minimizer at $x^* = 1$ that is not global (Exercise 0.2).

Proof

Proof. Let x^* be a local minimizer of f , so there is a $\delta > 0$ with $f(x^*) \leq f(x)$ whenever $\|x - x^*\| < \delta$ (Definition 0.6). Suppose x^* is not a global minimizer (Definition 0.5). Then some $y \in \mathbb{R}^p$ has $f(y) < f(x^*)$, and in particular $y \neq x^*$.

Let $t = \min \left\{ \frac{1}{2}, \frac{\delta}{2\|y - x^*\|} \right\}$, so $t \in (0, 1)$, and let $z = ty + (1-t)x^*$. Then $z - x^* = t(y - x^*)$, so $\|z - x^*\| = t\|y - x^*\| \leq \delta/2 < \delta$.

By convexity,

$$f(z) \leq tf(y) + (1-t)f(x^*) < tf(x^*) + (1-t)f(x^*) = f(x^*),$$

where the strict inequality uses $t > 0$ and $f(y) < f(x^*)$. So $f(z) < f(x^*)$ with $\|z - x^*\| < \delta$, which contradicts x^* being a local minimizer. Hence x^* is a global minimizer. $\square$

Infimum and supremum

Definition 0.8 (Infimum (greatest lower bound)). Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, meaning that some $t \in \mathbb{R}$ satisfies $t \leq a$ for all $a \in A$. The **infimum** of A , written $\inf A$, is the greatest real number t satisfying $t \leq a$ for all $a \in A$:

$$\inf A \stackrel{\text{def}}{=} \max\{t \in \mathbb{R} : \forall a \in A, t \leq a\}$$

If A is nonempty but not bounded below, we write $\inf A = -\infty$ by convention.

The maximum in Definition 0.8 always exists: that is the completeness (greatest-lower-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the infimum belongs to A , it equals the minimum: $\inf A = \min A$.

Example 0.1 (Numerical examples of infimum).

- $\inf\{1, 2, 3\} = 1$, since 1 is the smallest element.
- $\inf(0.5, 1] = 0.5 = \min[0.5, 1]$: for intervals open below, the infimum equals the minimum of the corresponding closed-below interval, even though $0.5 \notin (0.5, 1]$. More generally, $\inf(c, b] = \min[c, b] = c$ for any $c < b$.
- $\inf\{t \geq 0 : t > 0.5\} = 0.5$, even though 0.5 itself is not in the set.
- $\inf\{-1, -2, -3, \dots\} = -\infty$, because no real number is less than or equal to every element of that set.

Definition 0.9 (Supremum (least upper bound)). Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, meaning that some $t \in \mathbb{R}$ satisfies $a \leq t$ for all $a \in A$. The **supremum** of A , written $\sup A$, is the smallest real number t satisfying $a \leq t$ for all $a \in A$:

$$\sup A \stackrel{\text{def}}{=} \min\{t \in \mathbb{R} : \forall a \in A, a \leq t\}$$

If A is nonempty but not bounded above, we write $\sup A = +\infty$ by convention.

The minimum in Definition 0.9 always exists: that is the completeness (least-upper-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the supremum belongs to A , it equals the maximum: $\sup A = \max A$.

Example 0.2 (Numerical examples of supremum).

- $\sup\{1, 2, 3\} = 3$, since 3 is the largest element.
- $\sup\{t \geq 0 : t < 0.5\} = 0.5$, even though 0.5 itself is not in the set.
- $\sup\{1, 2, 3, \dots\} = +\infty$, because no real number is greater than or equal to every element of that set.

Sums

Theorem 0.8 (adding zero changes nothing).

$$a + 0 = a$$

Theorem 0.9 (Sums are symmetric).

$$a + b = b + a$$

Theorem 0.10 (Sums are associative).

When summing three or more terms, the order in which you sum them does not matter:

$$(a + b) + c = a + (b + c)$$

Products

Theorem 0.11 (Multiplying by 1 changes nothing).

$$a \times 1 = a$$

Theorem 0.12 (Products are symmetric).

$$a \times b = b \times a$$

Theorem 0.13 (Products are associative).

$$(a \times b) \times c = a \times (b \times c)$$

Division

Theorem 0.14 (Division can be written as a product). *If $b \neq 0$, then*

$$\frac{a}{b} = a \times \frac{1}{b}$$

Sums and products together

Theorem 0.15 (Multiplication is distributive).

$$a(b + c) = ab + ac$$

Exercise 0.4 (Expand a squared sum). Is $(3 + 4)^2$ equal to $3^2 + 4^2$? Then expand $(a + b)^2$ for any numbers a and b , using only the distributive law and the rules above.

Solution

Solution 0.4. No: $(3 + 4)^2 = 7^2 = 49$, while $3^2 + 4^2 = 9 + 16 = 25$. The difference, $49 - 25 = 24$, is $2 \cdot 3 \cdot 4$.

To see where that extra term comes from, write the square as a product and apply the distributive law (Theorem 0.15) twice:

$$\begin{aligned}(a + b)^2 &= (a + b)(a + b) && \text{(definition of a square)} \\ &= (a + b)a + (a + b)b && \text{(distributive law)} \\ &= (a^2 + ba) + (ab + b^2) && \text{(distributive law, twice)} \\ &= a^2 + ab + ab + b^2 && \text{(commutative and associative laws)} \\ &= a^2 + 2ab + b^2 && \text{(collect like terms)}\end{aligned}$$

The step “commutative and associative laws” uses Theorem 0.12 to write ba as ab , and Theorem 0.10 to drop the parentheses.

Theorem 0.16 (Square of a sum). *For any numbers a and b ,*

$$(a + b)^2 = a^2 + 2ab + b^2$$

Proof

Proof. By Solution 0.4. □

Replacing b by $-b$ gives $(a - b)^2 = a^2 - 2ab + b^2$. Squared errors such as $(y - \hat{y})^2$ are expanded this way.

Summation notation

Exercise 0.5 (Expand a sum). The expression $\sum_{i=1}^4 i^2$ is shorthand for a sum of four terms.

1. Guess which four terms, and add them up.
2. Write $\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$ for $N = 3$ without the $\sum$ symbol.

Solution

Solution 0.5.

1. Replace i by each of 1, 2, 3, 4 in turn, and add the results:

$$\begin{aligned}\sum_{i=1}^4 i^2 &= 1^2 + 2^2 + 3^2 + 4^2 \\ &= 1 + 4 + 9 + 16 \\ &= 30\end{aligned}$$

2. Replace i by each of 1, 2, 3 in turn inside the parentheses, add the three squares, and multiply the total by $\frac{1}{3}$:

$$\frac{1}{3} \sum_{i=1}^3 (y_i - \hat{y}_i)^2 = \frac{1}{3} [(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2]$$

Definition 0.10 (Summation notation). Let m and n be integers with $m \leq n$, and let $a_m, a_{m+1}, \dots, a_n$ be numbers. The **sum** of a_m through a_n is

$$\sum_{i=m}^n a_i \stackrel{\text{def}}{=} a_m + a_{m+1} + \dots + a_n$$

The variable i is the **index** of the sum; m and n are its **lower** and **upper limits**.

The index is a placeholder: $\sum_{i=1}^n a_i$ and $\sum_{j=1}^n a_j$ are the same number. When the upper limit is less than the lower limit, the sum has no terms, and by convention it equals 0.

Exercise 0.6 (Rearrange a sum). Let c be a number, and let a_1, a_2, a_3 and b_1, b_2, b_3 be numbers. Using Definition 0.10 and the rules of algebra above, show that:

1. $\sum_{i=1}^3 c a_i = c \sum_{i=1}^3 a_i$;

$$2. \sum_{i=1}^3 (a_i + b_i) = \sum_{i=1}^3 a_i + \sum_{i=1}^3 b_i.$$

Does either argument depend on there being exactly three terms?

Solution

Solution 0.6. Each step below applies Definition 0.10, the distributive law (Theorem 0.15), or the commutative and associative laws of addition (Theorem 0.9 and Theorem 0.10).

1. Expand the sum, then factor out c :

$$\begin{aligned} \sum_{i=1}^3 c a_i &= c a_1 + c a_2 + c a_3 && \text{(expand the sum)} \\ &= c (a_1 + a_2 + a_3) && \text{(distributive law)} \\ &= c \sum_{i=1}^3 a_i && \text{(collect the sum)} \end{aligned}$$

2. Expand the sum, then regroup the terms:

$$\begin{aligned} \sum_{i=1}^3 (a_i + b_i) &= (a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) && \text{(expand the sum)} \\ &= (a_1 + a_2 + a_3) + (b_1 + b_2 + b_3) && \text{(commutative and associative laws)} \\ &= \sum_{i=1}^3 a_i + \sum_{i=1}^3 b_i && \text{(collect the sum)} \end{aligned}$$

Neither argument uses the number of terms: the same steps work for any lower and upper limits.

Theorem 0.17 (A constant factor comes out of a sum). *For any number c and numbers $a_m, \dots, a_n$,*

$$\sum_{i=m}^n c a_i = c \sum_{i=m}^n a_i$$

Proof

Proof. By Solution 0.6, part 1. □

Theorem 0.18 (A sum of sums splits). For any numbers $a_m, \dots, a_n$ and $b_m, \dots, b_n$,

$$\sum_{i=m}^n (a_i + b_i) = \sum_{i=m}^n a_i + \sum_{i=m}^n b_i$$

i Proof

Proof. By Solution 0.6, part 2. □

Exercise 0.7 (Sum over a set). Let $A = \{-1, 2, 5\}$ and let $f(x) = x^2$. The expression $\sum_{x \in A} f(x)$ means: add up $f(x)$ for each element x of A .

1. Write $\sum_{x \in A} f(x)$ without the $\sum$ symbol, and compute its value.
2. List the elements of A in a different order, and add up the same terms in that order. Is the total the same?

Solution

Solution 0.7.

1. Replace x by each element of A in turn, and add the results:

$$\begin{aligned} \sum_{x \in A} f(x) &= f(-1) + f(2) + f(5) \\ &= (-1)^2 + 2^2 + 5^2 \\ &= 1 + 4 + 25 \\ &= 30 \end{aligned}$$

2. In the order $5, -1, 2$, the terms are 25, 1 and 4. Their total is $25 + 1 + 4 = 30$. The total is the same, because addition does not depend on the order of the terms.

Definition 0.11 (Sum over a finite set). Let $A = \{x_1, x_2, \dots, x_k\}$ be a finite set with $k \geq 1$ different elements. Let f be a function that gives a number $f(x)$ for each element x of A . The **sum of f over A** is

$$\sum_{x \in A} f(x) \stackrel{\text{def}}{=} f(x_1) + f(x_2) + \dots + f(x_k)$$

If A has no elements, the sum has no terms, and by convention it equals 0. For example, $\sum_{x \in \{\}} x^2 = 0$.

Remark 0.1 (The order of the terms does not matter). The order in which we list the elements of A does not change the sum. The reason is that addition is commutative and associative: we can reorder and regroup the terms of a finite sum without changing the total. For example, if $A = \{1, 2, 3\}$ and $f(x) = x^2$, listing A as 1, 2, 3 gives $1 + 4 + 9 = 14$, and listing A as 3, 1, 2 gives $9 + 1 + 4 = 14$. Exercise 0.7, part 2, shows another example. When $A = \{m, m + 1, \dots, n\}$, this sum is the same as $\sum_{i=m}^n f(i)$ from Definition 0.10.

Remark 0.2 (Leaving the set out). Some authors leave the set out and write $\sum_x f(x)$. This shorthand means the sum over every value x can take. For example, if x is the outcome of one roll of a six-sided die, $\sum_x f(x)$ means $\sum_{x \in \{1, 2, 3, 4, 5, 6\}} f(x)$. In these notes, we write the set out in full.

Quotients

Definition 0.12 (Quotient). For real numbers a and b with $b \neq 0$, the **quotient** of a by b is the result of dividing a by b :

$$\frac{a}{b}$$

A quotient is also called a *fraction*; a is its *numerator* and b its *denominator*. A quotient whose denominator measures time or population size is often called a *rate*; in epidemiology, rates typically have such a denominator.

cf. [https://en.wikipedia.org/wiki/Rate_\(mathematics\)](https://en.wikipedia.org/wiki/Rate_(mathematics))

Example 0.3 (A quotient). The quotient of 6 by 4 is $\frac{6}{4} = 1.5$. The quotient of 6 by 0 is undefined, because Definition 0.12 requires a nonzero denominator.

Definition 0.13 (Ratios). A **ratio** is a quotient in which the numerator and denominator are measured using the same unit scales.

cf. <https://en.wikipedia.org/wiki/Ratio>

Definition 0.14 (Proportion). In statistics, a **proportion** typically means a ratio where the numerator represents a subset of the denominator.

See https://en.wikipedia.org/wiki/Population_proportion.

See also [https://en.wikipedia.org/wiki/Proportion_\(mathematics\)](https://en.wikipedia.org/wiki/Proportion_(mathematics)) for other meanings.

Definition 0.15 (Proportional). Two functions $f(x)$ and $g(x)$ are **proportional** if their ratio $\frac{f(x)}{g(x)}$ does not depend on x . (cf. [https://en.wikipedia.org/wiki/Proportionality_\(mathematics\)](https://en.wikipedia.org/wiki/Proportionality_(mathematics)))

Additional reference for elementary algebra: https://en.wikipedia.org/wiki/Population_proportion#Mathematical_definition

Polynomials

Exercise 0.8 (Reading a polynomial). Let $f(x) = 3x^4 - x + 7$.

1. Write f in the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, listing every coefficient a_4, a_3, a_2, a_1, a_0 .
2. What is the degree of f , and what is its leading coefficient?

Solution

Solution 0.8.

1. $f(x) = 3x^4 + 0 \cdot x^3 + 0 \cdot x^2 + (-1) \cdot x + 7$, so $a_4 = 3$, $a_3 = 0$, $a_2 = 0$, $a_1 = -1$, and $a_0 = 7$. Powers of x that do not appear have coefficient 0.
2. The highest power with a nonzero coefficient is x^4 , so the degree is 4 and the leading coefficient is $a_4 = 3$.

Definition 0.16 (Polynomial). A **polynomial** (in one variable x) is a **function** $f : \mathbb{R} \rightarrow \mathbb{R}$ that can be written as

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

for some integer $n \geq 0$ and constants $a_0, a_1, \dots, a_n \in \mathbb{R}$ with $a_n \neq 0$.

In Exercise 0.8, $f(x) = 3x^4 - x + 7$ is a polynomial with $n = 4$, $a_4 = 3$, $a_3 = 0$, $a_2 = 0$, $a_1 = -1$, and $a_0 = 7$. A constant function $f(x) = 7$ is a polynomial with $n = 0$ and $a_0 = 7$. The requirement $a_n \neq 0$ means this definition covers nonzero polynomials only, so that the degree is defined (Definition 0.17); the zero function $f(x) = 0$ is excluded here.

Definition 0.17 (Degree of a polynomial). Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a **polynomial** with $a_n \neq 0$. The integer n is the **degree** of f .

In Exercise 0.8, $f(x) = 3x^4 - x + 7$ has degree 4. A constant polynomial $f(x) = 7$ has degree 0.

Definition 0.18 (Leading coefficient). Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial of degree n . The constant a_n is the **leading coefficient** of f .

In Exercise 0.8, $f(x) = 3x^4 - x + 7$ has leading coefficient 3.

Exponentials and Logarithms

In these notes, $\log\{x\}$ is the natural logarithm of $x > 0$, the logarithm with base $e \approx 2.718$, and $\exp\{x\} = e^x$ is the exponential function. Some sources write $\ln x$ for the natural logarithm and reserve $\log x$ for base 10.

Theorem 0.19 ($\exp\{\}$ and $\log\{\}$ are mutual inverses).

1. For every $a > 0$: $\exp\{\log\{a\}\} = a$.
2. For every $a \in \mathbb{R}$: $\log\{\exp\{a\}\} = a$.

Theorem 0.20 (Logarithm of a product). If $a > 0$ and $b > 0$, then

$$\log\{a \cdot b\} = \log\{a\} + \log\{b\}$$

Corollary 0.1 (Logarithm of a quotient). If $a > 0$ and $b > 0$, then

$$\log\left\{\frac{a}{b}\right\} = \log\{a\} - \log\{b\}$$

Proof

Proof. Since $a > 0$ and $b > 0$, the quotient $\frac{a}{b}$ is positive, so Theorem 0.20 applies to the product $\frac{a}{b} \cdot b$:

$$\begin{aligned} \log\{a\} &= \log\left\{\frac{a}{b} \cdot b\right\} && (a = \frac{a}{b} \cdot b) \\ &= \log\left\{\frac{a}{b}\right\} + \log\{b\} && (\text{logarithm of a product}) \end{aligned}$$

The second step applies Theorem 0.20. Subtracting $\log\{b\}$ from both sides gives $\log\{a\} - \log\{b\} = \log\{\frac{a}{b}\}$. $\square$

Theorem 0.21 (Logarithm of a power). *If $a > 0$ and $b \in \mathbb{R}$, then*

$$\log\{a^b\} = b \cdot \log\{a\}$$

Theorem 0.22 (exponential of a sum).

The exponential of a sum is equal to the product of the exponentials of the addends:

$$\exp\{a + b\} = \exp\{a\} \cdot \exp\{b\}$$

Corollary 0.2 (exponential of a difference).

The exponential of a difference is the exponential of the first term divided by the exponential of the second term:

$$\exp\{a - b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

Theorem 0.23 (Power of a power). *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$a^{bc} = (a^b)^c = (a^c)^b$$

Example 0.4 (A negative base). With $a = -1$, $b = 2$, and $c = \frac{1}{2}$:

$$\begin{aligned} a^{bc} &= (-1)^{2 \cdot \frac{1}{2}} \\ &= (-1)^1 \\ &= -1 \end{aligned}$$

but

$$\begin{aligned} (a^b)^c &= ((-1)^2)^{\frac{1}{2}} \\ &= 1^{\frac{1}{2}} \\ &= 1 \end{aligned}$$

So $a^{bc} \neq (a^b)^c$ here, which is why Theorem 0.23 requires $a > 0$. The third expression, $(a^c)^b = ((-1)^{\frac{1}{2}})^2$, is not even a real number.

Corollary 0.3 (natural exponential of a product).

$$\exp\{ab\} = (\exp\{a\})^b = (\exp\{b\})^a$$

Exercise 0.9. For $b, c \in \mathbb{R}$, when does $b^c = bc$?

Solution

Solution 0.9. We only count a pair (b, c) when b^c is a real number, so for $b < 0$ we only consider integer c (R agrees: $(-8)^{(1/3)}$ is NaN). With that convention, $bc = b^c$ in each of the following cases:

1. $c = 1$, for every b .
2. $b = 0$ and $c > 0$, since then $b^c = 0 = bc$. ($b = 0$ and $c = 0$ fails, since $0^0 = 1 \neq 0$.)
3. $b > 0, c > 0, c \neq 1$, and $b = \exp\left\{\frac{\log\{c\}}{c-1}\right\}$. For $b > 0$, dividing both sides of $b^c = bc$ by b gives $b^{c-1} = c$, which needs $c > 0$ because $b^{c-1} > 0$; taking logarithms then gives $(c-1)\log\{b\} = \log\{c\}$. For example, $c = 2$ gives $b = 2$, and indeed $2^2 = 4 = 2 \cdot 2$.
4. $b < 0$ and c is an odd integer with $c \geq 3$, with $b = -c^{1/(c-1)}$; for example, $b = -\sqrt[3]{3}$ and $c = 3$ give $b^c = -3\sqrt[3]{3} = bc$.
5. $b < 0$ and c is an even integer with $c \leq -2$, with $b = -(-c)^{1/(c-1)}$; for example, $b = -2^{-1/3}$ and $c = -2$ give $b^c = 2^{2/3} = bc$.

For $b < 0$, cases 4 and 5 come from $b^{c-1} = c$ as well: when $c-1$ is even, $b^{c-1} > 0$, so c must be positive; when $c-1$ is odd, $b^{c-1} < 0$, so c must be negative.

See the red contours in Figure 2 for a visualization of the $b \geq 0$ cases.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 1: Graph of $b * c$ and b^c

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 2: **Graph of $b^c - b * c$.** Red contour lines show where $b^c = b * c$.

Exercise 0.10. For $a \geq 0, b, c \in \mathbb{R}$, when does $(a^b)^c = a^{(b^c)}$?

Solution

Solution 0.10. Short answer: rarely (that's all you need to know for this course).

Long answer:

Split on whether $a > 0$ or $a = 0$, because the logarithm we use for $a > 0$ is undefined at $a = 0$.

Case $a > 0$. By Theorem 0.23, $(a^b)^c = a^{bc}$, so the question becomes when $a^{bc} = a^{(b^c)}$ (for pairs (b, c) where b^c is defined). Because $a > 0$, both sides are positive, and we can

take logarithms (Theorem 0.21):

$$\begin{aligned}
 a^{bc} &= a^{(b^c)} \\
 \log\{a^{bc}\} &= \log\{a^{(b^c)}\} \quad (\text{take logarithms of both sides}) \\
 bc \cdot \log\{a\} &= b^c \cdot \log\{a\} \quad (\text{logarithm of a power})
 \end{aligned} \tag{1}$$

The last line of Equation 1 holds exactly when

1. $a = 1$ (so that $\log\{a\} = 0$), or
2. $bc = b^c$ (see Exercise 0.9).

Case $a = 0$. Here we cannot take logarithms, so we work from the values of powers of 0: $0^s = 0$ for $s > 0$, $0^0 = 1$, and 0^s is undefined for $s < 0$.

- If $b < 0$, then 0^b is undefined, so $(a^b)^c$ is undefined.
- If $b > 0$, then $(0^b)^c = 0^c$ and $b^c > 0$, so $0^{(b^c)} = 0$; the two sides agree exactly when $c > 0$.
- If $b = 0$, then $(0^0)^c = 1^c = 1$; for $c > 0$, $0^{(0^c)} = 0^0 = 1$, so the two sides agree, and for $c \leq 0$ they do not.

So for $a = 0$, $(a^b)^c = a^{(b^c)}$ exactly when $b \geq 0$ and $c > 0$.

In particular, when $a = 0$, $b \geq 0$, and $c = 0$, the two sides differ:

$$\begin{aligned}
 (a^b)^c &= (0^b)^0 \\
 &= 1 \\
 a^{(b^c)} &= 0^{(b^0)} \\
 &= 0^1 \\
 &= 0
 \end{aligned}$$

Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.